CIS 220 Database Concepts
Case Studies
Concept: Discovery/Needs assessment/Conceptual ERD

In this document you will find a collection of case studies that are structured in the following way

- A narrative containing simple descriptions of the needs and goals of a ***proposed system***

- A "sample form" that is used to collect and track the data under question. These forms are generally ***master-detail*** type forms. A master–detail form is a common structure for depicting one-to-many type relationships, with one main entity and a collection of areas for associating multiple records of "detail". There may be multiple entities depicted in these forms and this could lead to multiple 1:M relationships.

- Due to the natural vagaries of language, you may need to make some assumptions about these problems. **If you do need to make an assumption, write that down as part of the analysis**

- Your task is to complete a discovery analysis for each case study, culminating in a ***conceptual Entity Relationship Diagram***. This diagram should be similar in form to the example provided in this document and the examples completed in class. Here are the requirements

    - ***Only entities*** are to be shown in the diagram, attributes will just be listed. The goal of these exercises is to focus solely on entities and relationship cardinality.

    Do not include attributes in the diagram. The conceptual diagram only focuses on entities and relationships.

    - Relationship cardinality notation (1:M, 1:1, M:N), minimum (optional/mandatory) and maximum must match the details of the case study and any assumptions you have noted
    - M:N relationships are **OK** in this diagram but must be noted. We will begin resolving these relationships next week when build ***Enhanced Entity Relationship Diagrams***
    - For each relationship shown in the diagram I want four plain language statements that describes the ***business rule*** that is being modeled. Check the example.

- I will include a sample of the above tasks for one case study. Then I will ask you to do three on your own. Use this sample to aid you in your brainstorming.

Sample: Rental Car Agency, Vehicle Maintenance

A small car rental agency wants to track its vehicles—particularly in terms of maintenance. Many of the customers rent the cars for a few weeks at a time and the company wants to ensure that routine maintenance is handled on a regular basis—such as changing the oil every 3,000 miles. Sometimes the company has to call people and ask them to come in for the basic maintenance.

Vehicle VIN
Make, Model, Year, Color
Type (SUV, Sedan, Hatchback, Truck, Wagon)
Date Purchased Initial Miles
Standard maintenance interval (miles)

Date	Miles	Maintenance	Description and Comments

Customer Rental
Customer Lastname, Firstname
Phone, Email
Address, City, State, ZIP

Vehicle
Date Rented Miles
Est. Return Date
Rental Rate
Damage or Comments

Return Date Return Comments Return Miles

Based on the case study above, we could make the following statements

Entities:

- Vehicle, Customer, Rental, Maintenance

Attributes:

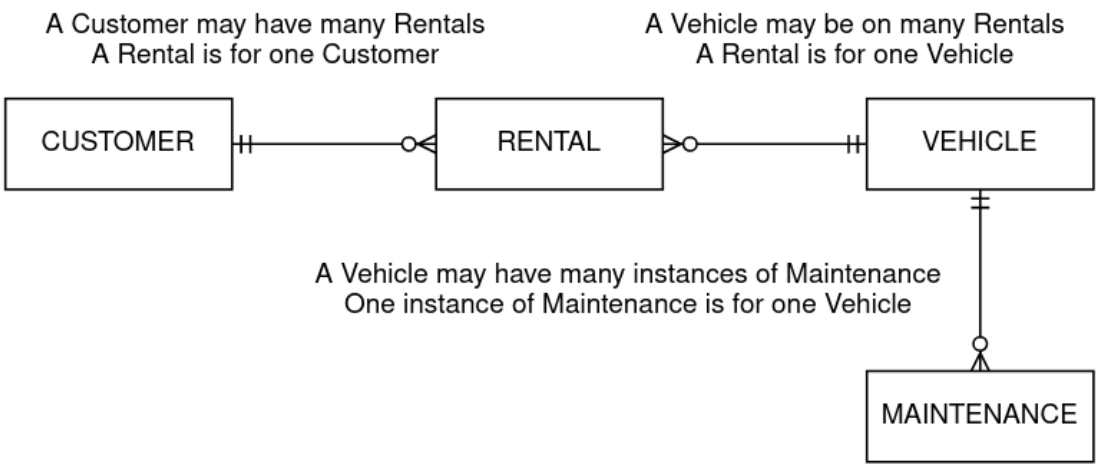
- **Customer:** First name, last name, phone, email, address, city, state, zip
- **Vehicle:** VIN, make, model, type, date purchased, initial miles, maintenance interval
- **Rental:** Vehicle, customer, date rented, estimated return date, rental rate, damage or comments, return date, return comments, return miles
- **Maintenance:** Vehicle, date, miles, description, comments

Relationships:

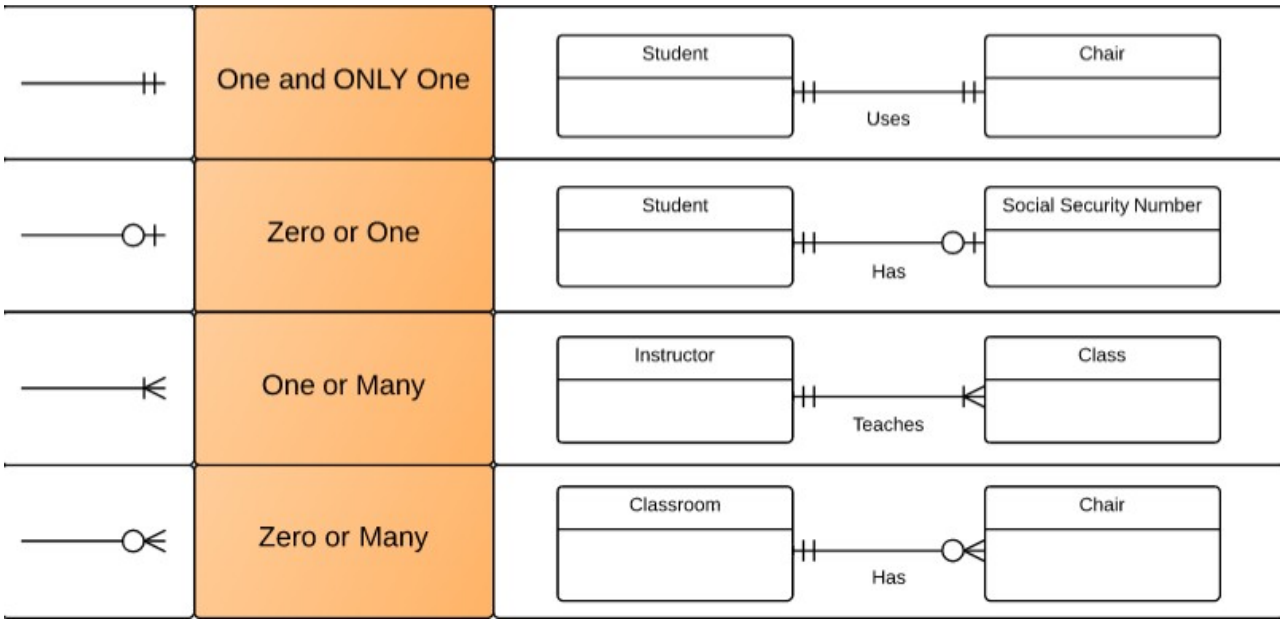
- **Assumption:** A Customer can have many Rentals. A Rental is for a single Customer
    - **Cardinality:** Customer 1:M Rental
    - **Multiplicity:**
        - A Customer may Rent (optional: min 0)
        - A Customer can Rent many times (max: many)
        - A Rental must be for a Customer (mandatory: min 1)
        - A Rental is only for a single Customer (max: 1)

- **Assumption:** A Rental agreement is for a single Car. A Car can be rented many times (be on many rental agreements) If a Customer wants to rent multiple vehicles, they will fill out one rental agreement for each vehicle.
    - **Cardinality:** Rental M:1 Car
    - **Multiplicity:**
        - A Rental agreement must be for a Vehicle (mandatory: min 1)
        - A Rental is for one Vehicle only (max: 1)
        - A Vehicle may be a part of a Rental Agreement (optional: min 0)
        - A Vehicle can be a part of many Rental Agreements (max: many)

- **Assumption:** A Vehicle can have many records of Maintenance. A Maintenance record only describes a single Vehicle.
    - **Cardinality:** Vehicle 1:M Maintenance
    - **Multiplicity:**
        - A Vehicle may have a Maintenance Record (optional: min 0)
        - A Vehicle can have many Maintenance Records (max many)
        - A Maintenance Record must be for a Vehicle (mandatory: min 1)
        - A Maintenance Record is for a single Vehicle only (max: 1)



ERD Crows's Foot Notation



Case Studies for your assignment begin on the next page. You are required to complete them all. There is no single correct answer to these case studies but there are some things that are objectively true. Be prepared to defend your design choices.

Property Assessment

You have been hired by a small county government office to track properties for tax purposes. The annual property tax rate is set by the local governments, but the assessor's office is responsible for tracking the value of each property. Whenever a house is sold, the sale price is reported to the office. Since this price is a concrete measure of the price, it has to be recorded permanently. In other cases, the assessor estimates the value of a property from its primary features, such as the lot size (measured in fractions of an acre), house size (square feet), age (year built), neighborhood, and general condition. New values are estimated every year if a house has not been sold. If a property has not sold for more than five years, an assessor performs a simple inspection of the property and takes pictures from the road. Property owners can appeal an assessment to a board. The appeal date and any comments are recorded. The data is currently recorded on a form organized by each property. Recently, the office has been assigning GPS coordinates (latitude, longitude, altitude) to each property. Many do not have data yet but will be added when the properties are inspected.

Property ID Year Built Year Remodel Address Tax Code Area Zone Code Subdivision	Construction Price Township/Range/ Section	Latitude Longitude Altitude
Lot size House size Sewer Water Utilities	Bedrooms Bathrooms Fireplaces Garage Basement	
Property Description		

Year	Valuation	Method	Comments
Appeals			
Date	Claimed Value	Resulting Value	Comments

Music Store

A local music store wants help with marketing. Specifically, the owner knows that some customers are better than others and wants to encourage them to purchase more items from the store by offering discounts. However, she does not want to offer the same level of discounts to everyone who walks in from the street because profits would drop. She needs a computer system to help employees give the correct discounts to each customer. Basically, she wants

a frequent buyer program, where customers are placed into categories based on the amount of money they spend in a year. However, she also wants the ability to move some customers into higher categories based on other factors (such as name recognition). The heart of the system is a list of the customer categories, the sales amount needed to reach each level, and the discount applied. However, some products or sales will not count towards the reward levels. For instance, close-out sales items will not be included in sales totals, nor are they eligible for additional discounts. So, she needs a list of inventory items along with an indicator for the discount eligibility.

Level	Sales	Discount
Platinum	$50,000	20%
Gold	$25,000	15%
Silver	$10,000	10%
Bronze	$5,000	5%

Sale ID					Sale Date Employee
Customer ID Last Name, First Name Band E-mail Address Phone Reward Level		Delivery Address City State, ZIP			
Item	Description	Price	Quantity	Discount	Value
Current annual total of reward points:					Total Reward Points

Document Translation

A friend of yours has a business that translates documents for other companies. She knows several languages and started out doing most of the translations herself. As her business grew, she found more people to work
as part-time translators. Most translators specialize in translating from one language into their native tongue. A few can translate from several languages. Several companies send documents for translation on a regular basis. Others are one-time translations. With more business and several translators, your friend needs to track which people are working on documents. The translators set the price for each translation, but pay a commission to your friend. The commission is generally 25 percent of the price, with a minimum of $5 per contract. The main data will be tracked through a Web site form that is filled in by the translators.

Contract ID					Contract Date		
Client Contact Phone Address, City, State/Province Country, Postal Code					Department E-mail		
Translator E-mail Native langauge					Phone (with country code)		
Language		Skill Level					
Documents			Target Language				
Name	Language	Date Received	Date Due	Initial Date	Review Date	Price	Difficulty
						Total	